THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 14: **Math of Space: Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The total surface area of a cuboid of length 8 cm, width 5 cm and height 3 cm is: 1

(a) 158 cm^2 **(b)** 120 cm^2 **(c)** 79 cm^2 **(d)** 89 cm^2

2. The volume of a cuboid measuring 7 cm $\times$ 6 cm $\times$ 5 cm is: 1

(a) 18 cm^3 **(b)** 230 cm^3 **(c)** 214 cm^3 **(d)** 210 cm^3

3. The volume of a cube is 64 cm^3. Its total surface area is: 1

(a) 48 cm^2 **(b)** 16 cm^2 **(c)** 64 cm^2 **(d)** 96 cm^2

4. Taking $\pi = \frac{22}{7}$, the curved surface area of a cylinder of radius 14 cm and height 6 cm is: 1

(a) 528 cm^2 **(b)** 264 cm^2 **(c)** 3696 cm^2 **(d)** 550 cm^2

5. Taking $\pi = \frac{22}{7}$, the curved surface area of a cone of base radius 7 cm and slant height 25 cm is: 1

(a) 704 cm^2 **(b)** 550 cm^2 **(c)** 1100 cm^2 **(d)** 275 cm^2

6. A cone has base radius 6 cm and height 8 cm. Its slant height is: 1

(a) 9 cm **(b)** 10 cm **(c)** 11 cm **(d)** 14 cm

7. Taking $\pi = \frac{22}{7}$, the volume of a sphere of radius 21 cm is: 1

(a) 38808 cm^3 **(b)** 29106 cm^3 **(c)** 19404 cm^3 **(d)** 77616 cm^3

8. Taking $\pi = \frac{22}{7}$, the volume of a hemisphere of radius 21 cm is: 1

(a) 19558 cm^3 **(b)** 38808 cm^3 **(c)** 9702 cm^3 **(d)** 19404 cm^3

9. The surface area of a cylindrical tumbler closed at one end and open at the other (radius r, height h) is: 1

(a) $\pi r(h + 2r)$ **(b)** $\pi r(2h + r)$ **(c)** $2\pi r(h + r)$ **(d)** $2\pi rh$

10. **Assertion (A):** The volume of a cone is $\frac{1}{3}\pi r^2 h$. 1

Reason (R): Three cones of the same base and height fill exactly one cylinder of that base and height.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The volume of a cube is 343 cm^3. Find its side and total surface area. 2

12. Find the volume of a water tank measuring 2.5 m $\times$ 2 m $\times$ 1.2 m. How many litres does it hold? 2

13. Find the curved surface area of a cylinder of radius 7 cm and height 20 cm ($\pi = \frac{22}{7}$). 2

14. A cone has height 24 cm and base radius 7 cm. Find its slant height. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of side 6 cm is cut into cubes of side 1 cm. Find the ratio of the surface area of the original cube to the total surface area of all the small cubes. 3

16. A godown measures 30 m $\times$ 20 m $\times$ 8 m. How many cuboidal boxes of size 2 m $\times$ 1.25 m $\times$ 1 m can be stored in it? 3

17. A cube of side 4 cm is painted on all faces and then cut into 64 cubes of side 1 cm. How many small cubes have 3
(i) three faces painted,
(ii) two faces painted,
(iii) one face painted,
(iv) no face painted?

OR

A cylindrical drum has radius 0.7 m and height 1.5 m. Find the volume of oil it can hold in litres ($\pi = \frac{22}{7}$, $1\ \text{m}^3 = 1000$ litres). 3

18. The curved surface area of a cylinder is $264\ \text{cm}^2$ and its height is 7 cm. Find its radius and volume ($\pi = \frac{22}{7}$). 3

OR

The radii of two cylinders are in the ratio $1 : 2$ and their heights are in the ratio $4 : 3$. Find the ratio of 3
(i) their volumes,
(ii) their curved surface areas.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Derive the formulas for the curved surface area and total surface area of a cylinder of radius r and height h. Hence find both for $r = 7$ cm and $h = 15$ cm ($\pi = \frac{22}{7}$). 5

OR

Case study: A cylindrical water tank has radius 1.4 m and height 3 m ($\pi = \frac{22}{7}$). 5
(i) Find its capacity in litres.
(ii) Find the area of the curved surface to be painted.
(iii) Find the cost of painting at ₹25 per m^2.
(iv) A family uses 120 litres a day. For how many days does a full tank last?

20. A cone has base radius 7 cm and height 24 cm ($\pi = \frac{22}{7}$). 5
(i) Find its slant height.
(ii) Find its curved and total surface area.
(iii) Find its volume.
(iv) State how the volume compares with a cylinder of the same base and height.

OR

Case study: A conical tent has base radius 7 m and height 24 m ($\pi = \frac{22}{7}$). 5
(i) Find the slant height.
(ii) Find the canvas required.
(iii) Find the volume of air inside.
(iv) If each person needs $11\ \text{m}^3$ of air, how many people can sleep in the tent?

END OF QUESTION PAPER Total: 40 Marks